

# Single-Asset Geometric Brownian Motion

## LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Derive the GBM solution.** Apply Itô's lemma to obtain the exact lognormal solution and explain the  $-\sigma^2/2$  correction.
- **Estimate and simulate.** Map growth-rate moments into drift and volatility estimates and simulate exact discrete-time GBM paths.
- **Evaluate model limits.** Compute a target-ROI probability and test GBM against heavy growth-rate tails, weak growth-rate autocorrelation, and volatility clustering.

## 1 From Random Walks to Positive Prices

Brownian motion entered finance through attempts to describe continuously fluctuating prices. Louis Bachelier used an arithmetic Brownian model in 1900; later work associated with Paul Samuelson emphasized multiplicative dynamics so prices remain positive. Geometric Brownian motion (GBM) became foundational because it has an analytic solution, scales noise with price, and connects cleanly to option-pricing theory. These virtues make it a benchmark, not an empirical law.

GBM assumes the ex-dividend share price  $S_t > 0$  satisfies the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (1)$$

where  $\mu \in \mathbb{R}$  is the annual arithmetic drift in the price SDE,  $\sigma > 0$  is annual volatility with units  $\text{years}^{-1/2}$ , and  $W_t$  is a standard Wiener process. We reserve  $g$  for an observed continuously compounded growth rate. Both  $\mu$  and  $g$  have units  $\text{years}^{-1}$ , but they are not the same quantity.

### Definition 1.1. Standard Wiener process

A standard Wiener process  $(W_t)_{t \geq 0}$  is a continuous real-valued stochastic process satisfying  $W_0 = 0$  almost surely; increments over disjoint intervals are independent; and for  $0 \leq s < t$ , the increment  $W_t - W_s$  is normally distributed with mean zero and variance  $t - s$ . Consequently,

$$W_t - W_s \stackrel{d}{=} \sqrt{t-s} Z, \quad Z \sim \mathcal{N}(0, 1), \quad (2)$$

where  $\stackrel{d}{=}$  denotes equality in distribution.

Independent Gaussian increments give GBM tractability and also predetermine several of its empirical failures.

## 2 Itô's Lemma Creates the Variance Correction

Ordinary calculus would incorrectly write  $d \log S_t = dS_t/S_t$ . Brownian increments have quadratic variation of order  $dt$ , so the second derivative contributes at first order. Itô's lemma states the required stochastic chain rule ([Thm. 2.1](#)).

### Theorem 2.1. One-dimensional Itô formula

Let  $X_t$  satisfy  $dX_t = a(X_t, t) dt + b(X_t, t) dW_t$ , and let  $f(x, t)$  be once continuously differentiable in  $t$  and twice continuously differentiable in  $x$ . Under standard integrability conditions,  $Y_t := f(X_t, t)$  satisfies

$$dY_t = \left( \frac{\partial f}{\partial t} + a \frac{\partial f}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} \right) dt + b \frac{\partial f}{\partial x} dW_t. \quad (3)$$

All coefficients and derivatives on the right are evaluated at  $(X_t, t)$ .

Apply [Thm. 2.1](#) to  $f(S) = \log S$ . Because  $f'(S) = 1/S$  and  $f''(S) = -1/S^2$ , the GBM dynamics imply

$$d \log S_t = \left( \mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t, \quad (4)$$

$$\log \left( \frac{S_T}{S_0} \right) = \left( \mu - \frac{\sigma^2}{2} \right) T + \sigma W_T. \quad (5)$$

Exponentiation gives the exact GBM solution ([Prop. 2.1](#)).

### Proposition 2.1. Exact solution and distribution of GBM

For constant  $\mu \in \mathbb{R}$ ,  $\sigma > 0$ , initial price  $S_0 > 0$ , and horizon  $T > 0$  years, the solution of [Eq. 1](#) is

$$S_T = S_0 \exp \left[ \left( \mu - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} Z \right], \quad Z \sim \mathcal{N}(0, 1). \quad (6)$$

Thus  $\log(S_T/S_0)$  is normal with mean  $(\mu - \sigma^2/2)T$  and variance  $\sigma^2 T$ , while  $S_T$  is lognormal and strictly positive almost surely.

The correction  $-\sigma^2/2$  reconciles arithmetic and log growth. From the lognormal moments,

$$\mathbb{E}[S_T] = S_0 e^{\mu T}, \quad \text{Var}(S_T) = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1). \quad (7)$$

The median is  $S_0 e^{(\mu - \sigma^2/2)T}$ , below the mean when  $\sigma > 0$  because the price distribution is right-skewed.

### COMPANION NOTEBOOK

#### L4b: Derivation of the GBM Solution →

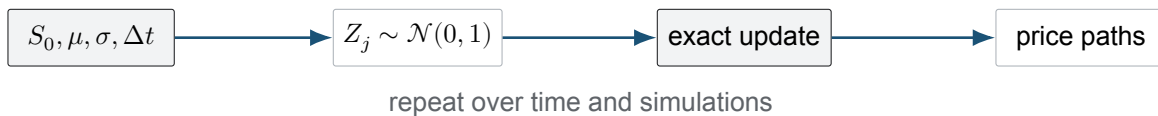
Work through Wiener increments, Itô's lemma, and the analytic transition from the SDE to the lognormal solution.

### 3 Exact Discrete Simulation

For a grid  $0 = t_0 < t_1 < \dots < t_N$  with step lengths  $\Delta t_j := t_j - t_{j-1} > 0$  years, GBM can be sampled exactly at grid points:

$$S_{t_j} = S_{t_{j-1}} \exp \left[ \left( \mu - \frac{\sigma^2}{2} \right) \Delta t_j + \sigma \sqrt{\Delta t_j} Z_j \right], \quad Z_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (8)$$

This is an exact transition for constant-parameter GBM, not the Euler–Maruyama approximation. Monte Carlo repeats Eq. 8 with independent pseudorandom streams to approximate path or terminal distributions. Reproducibility requires recording the seed and generator, while uncertainty in Monte Carlo estimates should be distinguished from uncertainty in  $\mu$  and  $\sigma$ .



**Figure 1:** Exact grid simulation of constant-parameter GBM. Gaussian shocks generate process uncertainty; parameter estimates and model form introduce separate layers of uncertainty.

### 4 Estimating Drift and Volatility Correctly

For equally spaced observations, define the growth rate  $g_j := (1/\Delta t) \log(S_{t_j}/S_{t_{j-1}})$  and its associated log return  $r_j := (\Delta t)g_j$ . Under GBM,

$$g_j \stackrel{\text{iid}}{\sim} \mathcal{N} \left[ \mu - \frac{\sigma^2}{2}, \frac{\sigma^2}{\Delta t} \right]. \quad (9)$$

Write  $\bar{g} := \mu - \sigma^2/2$  for the mean growth rate. If  $g'$  is the sample mean and  $\sigma_g$  is the sample standard deviation of the growth rates (the L3a risk measure), the natural moment estimates are

$$\hat{\sigma} = \sigma_g \sqrt{\Delta t}, \quad \hat{\bar{g}} = g', \quad \hat{\mu} = \hat{\bar{g}} + \frac{1}{2} \hat{\sigma}^2. \quad (10)$$

The sample mean  $g'$  estimates the mean growth rate  $\bar{g} = \mu - \sigma^2/2$ , not the arithmetic SDE drift itself, and  $\sigma_g$  estimates  $\sigma/\sqrt{\Delta t}$ , not  $\sigma$ . Omitting the half-variance adjustment, or the  $\sqrt{\Delta t}$  scaling, silently changes the meaning of the model parameter.

The drift parameter is difficult to estimate over short samples because expected growth is small relative to growth-rate noise. Regressing log price on time does not remove stochastic residuals and generally creates serially correlated regression errors. Likelihood or growth-rate moment methods align more directly with Eq. 9, but they remain vulnerable to changing regimes, outliers, selection choices, and model misspecification.

#### COMPANION NOTEBOOK

##### L4b: Estimating Single-Asset GBM Parameters →

Estimate the mean growth rate and volatility from historical prices, test the estimation assumptions, simulate paths, and compare exact lognormal quantile bands with an observed price trajectory.

## 5 Closed-Form Target Probability

Use the present-value return definition from L4a without costs, with  $g_b$  the continuously compounded annual benchmark growth rate:  $\rho_T := e^{-g_b T} S_T / S_0 - 1$ . For a target  $\rho_\star > -1$ , substitute Eq. 6 and standardize the normal shock. The success probability is

$$\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi\left(\frac{\log(1 + \rho_\star) - (\mu - g_b - \sigma^2/2)T}{\sigma\sqrt{T}}\right), \quad (11)$$

where  $\Phi$  is the standard normal cumulative distribution function and the probability uses the real-world parameters  $\mu$  and  $\sigma$ . Replacing  $\mu$  by a risk-neutral drift changes the measure and no longer answers an empirical probability-of-profit question.

### COMPANION NOTEBOOK

#### L4b: GBM NPV Trade Rule →

Evaluate the probability that a scheduled long position clears a target return under GBM, check the implementation against the median case, and plot the probability against the target.

## 6 Which Stylized Facts Does GBM Encode?

GBM growth rates over nonoverlapping equal-length intervals are independent Gaussian observations with constant variance rate. The model therefore produces zero autocorrelation in those growth rates, consistent with one broad stylized fact. It cannot generate heavy-tailed growth rates or volatility clustering because  $\sigma$  is constant and Gaussian tails are built in. The lognormal price distribution should not be confused with a heavy-tailed growth-rate law.

GBM can still be useful for short-horizon scenarios, benchmarking, and deriving analytical intuition. Its adequacy must be judged against the decision: a model used to estimate extreme loss or dynamic hedging error demands stronger tail and volatility validation than one used to teach stochastic calculus.

## Optional Advanced Material

---

### COMPANION NOTEBOOK

#### **L4b Advanced: From the Binomial Lattice to GBM** →

Match the binomial factors and probability to  $\mu$  and  $\sigma$ , and watch the terminal lattice distribution converge to the lognormal as the step count grows.

### COMPANION NOTEBOOK

#### **L4b Advanced: First-Passage Rules Under GBM** →

Compute the closed-form probability of reaching a take-profit level under GBM, evaluate two-barrier take-profit and stop-loss rules with the L4a absorbing recursion and Monte Carlo, and quantify the effect of the monitoring frequency.

### COMPANION NOTEBOOK

#### **L4b Advanced: Drift Uncertainty** →

Show that the drift estimate depends on the calendar span rather than the sampling frequency, and propagate that uncertainty into the target probability.

### COMPANION NOTEBOOK

#### **L4b Advanced: Monte Carlo Versus the Closed Form** →

Estimate the target probability by simulation with standard errors, and compare the exact transition with an Euler time-stepping scheme.

**KEY TAKEAWAYS**

In this lecture, we solved GBM with Itô's lemma, separated arithmetic drift from mean log growth, and derived an exact terminal target probability.

- **Quadratic variation matters.** The Itô correction  $-\sigma^2/2$  appears because Brownian squared increments contribute at order  $dt$ .
- **Drift names must be consistent.** The SDE uses arithmetic drift  $\mu$ ; the one-step growth rate  $g_j = (1/\Delta t) \log(S_{t_j}/S_{t_{j-1}})$  has mean  $\bar{g} = \mu - \sigma^2/2$  and standard deviation  $\sigma/\sqrt{\Delta t}$ .
- **Tractability has a cost.** Independent Gaussian increments give exact simulation and closed-form probabilities but exclude heavy tails and volatility clustering.

Next, couple several GBM assets with correlated shocks and generate portfolio weights from a Dirichlet distribution.

**ADDITIONAL READING**

- Bernt Øksendal, *Stochastic Differential Equations*, 6th ed. (2003), for Wiener processes, Itô's formula, and SDE solutions.
- John C. Hull, *Options, Futures, and Other Derivatives*, 11th ed. (2021), for lognormal price models and risk-neutral applications.
- Paul A. Samuelson, "Proof That Properly Anticipated Prices Fluctuate Randomly," *Industrial Management Review* 6(2), 41–49 (1965), for the modern economic random-walk argument.

**Educational use and risk notice.** These notes are for instruction only and do not constitute investment advice. Financial decisions involve risk, and model outputs depend on assumptions.